

Qns	Answer	Marks
1(a)	the change in length of a material is directly proportional to the force applied to it provided limit of proportionality is not exceeded	B1
1(b)	let F be the applied load and e be the extension $F = ke$ $k = \frac{F}{e} = \frac{1.0}{5.0 \times 10^{-2}}$ $= 20.0 \text{ N m}^{-1}$	M1
1(c)	[magnitude] for the same mass and spring used, the magnitude of acceleration of directly proportional to the vertical displacement of mass from its equilibrium position [direction] the negative sign shows that the direction of acceleration is always opposite to that of vertical displacement	B1 B1
1(d)	let displacement from natural unstretched length be x_{eqm} $W = kx_{\text{eqm}}$ $x_{\text{eqm}} = \frac{mg}{k} = 0.1 * 9.81 / 20$ $(= 0.04905 \text{ m})$ elastic potential energy $= \frac{1}{2} kx_{\text{eqm}}^2 = \frac{1}{2} k \frac{(mg)^2}{k^2} = \frac{(mg)^2}{2k}$ $= \frac{((0.10)(9.81))^2}{2(20)}$ $= 0.024 \text{ J}$	M1

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1(e)	<table><tr><th></th><th>GPE / J</th><th>EPE / J</th><th>KE / J</th><th>total E / J</th></tr><tr><td>lowest point</td><td>0</td><td>2.4</td><td>0</td><td>2.4</td></tr><tr><td>equilibrium position</td><td>1.2</td><td>0.60</td><td>0.6</td><td>2.4</td></tr><tr><td>highest point</td><td>2.4</td><td>0</td><td>0</td><td>2.4</td></tr></table>		GPE / J	EPE / J	KE / J	total E / J	lowest point	0	2.4	0	2.4	equilibrium position	1.2	0.60	0.6	2.4	highest point	2.4	0	0	2.4	B1 B1 B1
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EPE at lowest point: $\frac{1}{2}k(x_{\text{eqm}} + x)^2 = 2.4(059) \text{ J}$ GPE at eqm: $mgh = (0.50)(9.81)(x_0) = 1.2(030) \text{ J}$ GPE at highest point: $2mgx_0 = 2.4(059) \text{ J}$ KE at eqm: $\frac{1}{2}mv_{\text{max}}^2 = \frac{1}{2}m\omega^2x_0^2 = \frac{1}{2}m\frac{k}{m}x_0^2 = \frac{1}{2}kx_0^2 = 0.60 \text{ J}$																						
1(f)		B1 each graph																				
2(a)(i)	T ₁ and T ₂ : down	B1																				